

# Standard Deviation of a Standardized Vector

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September 2026

Assume  $\mathbf{x}$  is not a constant vector, so  $\text{std}(\mathbf{x}) \neq 0$ . Therefore, the vector  $\mathbf{z}$ , the *standardized* version of  $\mathbf{x}$ , is well defined.

$$\begin{aligned}\mathbf{z} &= \frac{\tilde{\mathbf{x}}}{\text{std}(\mathbf{x})} \\ &= \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}} \quad \because \text{std}(\mathbf{x}) = \frac{\|\tilde{\mathbf{x}}\|}{\sqrt{n}}\end{aligned}$$

Since  $\mathbf{z}$  is a positive scalar multiple of  $\tilde{\mathbf{x}}$ , and scaling preserves the average.

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It is easy to see why scaling preserves the average:

$$\begin{aligned}\mathbf{x} &= (x_1, x_2, \dots, x_n) \\ \text{avg}(\mathbf{x}) &= \frac{x_1 + x_2 + \dots + x_n}{n} \\ \alpha \mathbf{x} &= (\alpha x_1, \alpha x_2, \dots, \alpha x_n) \\ \text{avg}(\alpha \mathbf{x}) &= \frac{\alpha x_1 + \alpha x_2 + \dots + \alpha x_n}{n} \\ &= \frac{\alpha (x_1 + x_2 + \dots + x_n)}{n} \\ &= \alpha \text{avg}(\mathbf{x})\end{aligned}$$

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Therefore,

$$\begin{aligned}\text{avg}(\mathbf{z}) &= \text{avg}\left(\frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}}\right) \\ &= \left(\frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|}\right) \text{avg}(\tilde{\mathbf{x}}) \\ &= 0 \quad \because \text{avg}(\tilde{\mathbf{x}}) = 0\end{aligned}$$

Let us try to determine  $\text{std}(\mathbf{z})$ .

$$\begin{aligned}\text{std}(\mathbf{z}) &= \frac{\|\tilde{\mathbf{z}}\|}{\sqrt{n}} \\&= \frac{\|\mathbf{z}\|}{\sqrt{n}} && \because \text{avg}(\mathbf{z}) = 0 \\&= \frac{1}{\sqrt{n}} \left\| \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \tilde{\mathbf{x}} \right\| \\&= \frac{1}{\sqrt{n}} \cdot \frac{\sqrt{n}}{\|\tilde{\mathbf{x}}\|} \cdot \|\tilde{\mathbf{x}}\| && \because \|c \mathbf{v}\| = |c| \|\mathbf{v}\| \text{ for } c > 0 \\&= 1\end{aligned}$$

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